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BMMS2094 STATISTICS FOR DATA SCIENCE

GROUP ASSIGNMENT REPORT

Forecasting Battery-Electric Car Registrations in Singapore

Time-series evidence for sustainable transport planning

Dataset

Land Transport Authority: Monthly New Registration of Cars by Make

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Forecasting Battery-Electric Car Registrations in Singapore

A comparative R analysis of trend, smoothing and SARIMA models

ABSTRACT

This study forecasts monthly battery-electric car registrations in Singapore using current Land Transport Authority data. After restricting fuel type to Electric and excluding structural blank cells, the usable series contained 49 observations from 2022-07 to 2026-07. Four course-aligned forecasting models and a seasonal-naive benchmark were assessed on the final 12 months. Holt-Winters achieved the lowest RMSE (373.8) and MAE (311.7), while its Ljung-Box p-value of 0.931 indicated no detected residual autocorrelation at the tested lag. Forecasts suggest continued growth but widening uncertainty. The results can support charging and transport-capacity planning, but registrations are not direct measurements of emissions reductions.

Keywords - electric vehicles, forecasting, Holt-Winters, SARIMA, SDG 13.

I. INTRODUCTION

Singapore is accelerating the transition to cleaner-energy vehicles as part of its land-transport decarbonisation strategy [1]. Monthly registration data provide a timely indicator of market adoption and potential future demand for charging infrastructure. The forecasting question is therefore operational: how many new battery-electric cars may be registered over the next twelve months?

II. OBJECTIVES

The study aims to (1) construct a reproducible monthly battery-electric registration series; (2) describe its trend, seasonality and dependence; (3) compare four forecasting methods under a common holdout design; and (4) translate the forecast into cautious sustainability implications.

III. DATASET AND SDG ALIGNMENT

The August 2026 LTA archive contains 61,359 category rows from January 2016 to July 2026 [2]. A distinct Electric category is reported only from July 2022; earlier records were not relabelled as EVs. Blank numeric cells were treated as non-reported category combinations and excluded, not automatically converted to zero. The resulting 49-month series has no internal missing months. The analysis primarily supports SDG 13.2 by informing transport-decarbonisation planning [3], with SDG 11.6 as a secondary urban-environment link.

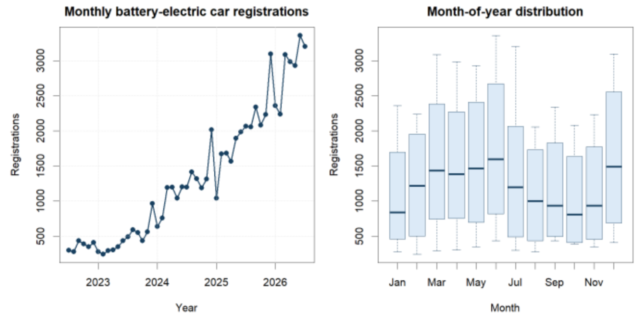

Fig. 1. Monthly electric registrations and month-of-year distribution.

IV. DESCRIPTIVE FINDINGS

Monthly registrations averaged 1,344.8, with a median of 1,197. Values ranged from 241 to 3,357. The standard deviation was 938.7, consistent with rapid market expansion and increasing variability. The short history limits the separation of stable annual seasonality from adoption growth and policy effects.

V. DATA PREPARATION

The R pipeline downloads and preserves the official ZIP, checks the six-field schema, parses monthly dates, audits duplicate composite keys, filters fuel_type == "Electric", removes blank numeric combinations, aggregates across make, importer and vehicle type, and checks calendar continuity. No composite-key duplicates or internal missing months were found.

VI. EXPERIMENTAL DESIGN

July 2022-July 2025 formed the training set and August 2025-July 2026 the untouched test set. Member 1 contributes trend-plus-season regression; Member 2 additive Holt-Winters; Member 3 ETS; and Member 4 Box-Jenkins SARIMA. Seasonal naive is the shared benchmark. Models are compared using MAE, RMSE, MAPE, MASE and sMAPE, followed by residual ACF and Ljung-Box checks.

ARIMA stabilisation: the rising EV level was handled inside the ARIMA candidate by one ordinary difference, Delta y(t) = y(t) - y(t-1). Auto-selection produced ARIMA(0,1,2) with drift: d = 1 removes the non-stationary level, the drift retains the average direction of growth, and D = 0 means that no seasonal difference was retained. This transformation applies to the ARIMA candidate only; Holt-Winters, ETS and regression use their own trend and seasonal structures.

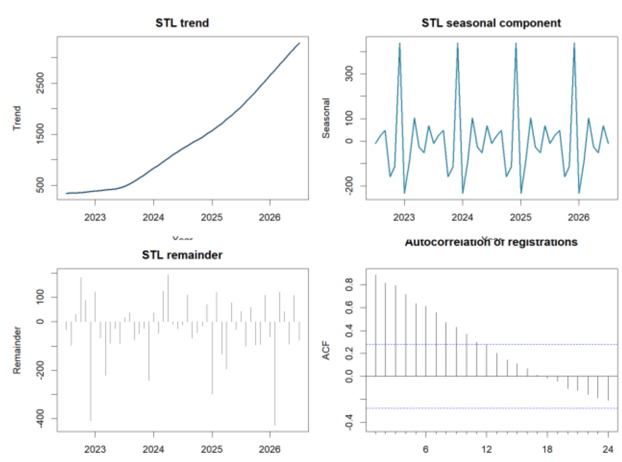

Fig. 2. STL decomposition and autocorrelation of the EV series.

VII. FORECAST ACCURACY

Model	Train MAE	Test MAE	Train RMSE	Test RMSE
Holt-Winters	133.2	311.7	181.2	373.8
SARIMA	141.3	333.9	188.9	424.8
Trend + season	146.4	513.0	166.1	583.6
ETS	0.2	656.9	0.3	804.3
Seasonal naive	603.2	1068.8	664.3	1102.6

TABLE I. Training residual accuracy versus untouched 12-month test accuracy.

Holt-Winters ranked first with RMSE 373.8, MAE 311.7, MAPE 10.78% and MASE 0.517. Its training RMSE was 181.2 versus test RMSE 373.8. SARIMA was second on the test set (RMSE 424.8). Training values are in-sample residual errors, whereas test values are genuine unseen-period forecast errors; therefore the model ranking uses test RMSE. The seasonal-naive benchmark produced test RMSE 1102.6, so the selected model substantially improved on repeating last year's values.

VIII. DIAGNOSTICS

The selected model's Ljung-Box statistic was 1.88 with p = 0.931. At the chosen lag, the null hypothesis of uncorrelated residuals was not rejected. Trend-plus-season and seasonal-naive residuals retained significant dependence, weakening their suitability despite interpretability.

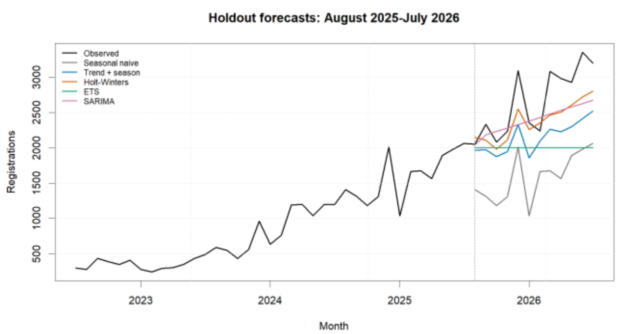

Fig. 3. Training history and common 12-month test forecasts. The dashed line marks the train-test split; coloured lines are model forecasts and black is observed data.

IX. TWELVE-MONTH FORECAST

Month	Forecast	95% interval
2026-08	3,427	3,030-3,824
2026-09	3,508	3,095-3,922
2026-10	3,372	2,927-3,816
2026-11	3,541	3,051-4,031
2026-12	4,166	3,615-4,717
2027-01	3,643	3,016-4,269
2027-02	3,818	3,104-4,531
2027-03	4,213	3,401-5,025
2027-04	4,195	3,276-5,115
2027-05	4,280	3,243-5,316
2027-06	4,508	3,347-5,669
2027-07	4,531	3,238-5,823

TABLE II. Holt-Winters forecasts, rounded to registrations.

The point forecast rises from approximately 3,427 registrations in August 2026 to 4,531 in July 2027. Month-to-month variation remains visible, while intervals widen with horizon. These are conditional statistical projections rather than policy targets.

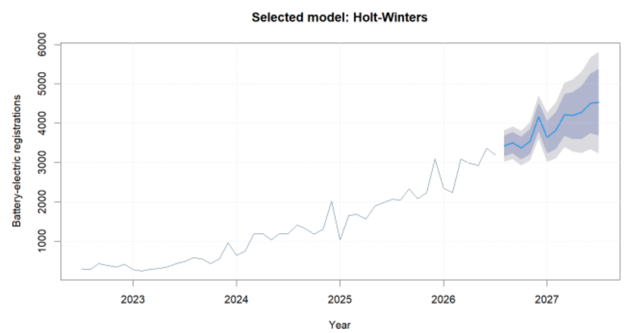

Fig. 4. Selected-model forecast with 80% and 95% intervals.

X. DISCUSSION

The evidence is consistent with a rapidly expanding battery-electric car market. Forecast growth can help agencies and operators stress-test charging-point rollout, electrical capacity, servicing and registration workflows. The strong result for Holt-Winters suggests that locally evolving level, trend and seasonal effects captured the holdout better than a fixed seasonal repeat or a nonseasonal ETS specification.

XI. LIMITATIONS

Only 49 monthly EV observations were available, leaving three years of training data after reserving a 12-month test. Structural change, incentives, COE conditions, model availability and COVID-era recovery are not causal regressors. LTA registrations exclude taxis and measure new registrations rather than fleet usage, electricity source, lifecycle emissions or avoided emissions. Prediction intervals reflect model uncertainty, not every policy or supply shock.

XII. RECOMMENDATIONS

Update the model monthly, monitor residual drift, and compare forecasts with charging-point utilisation and cleaner-energy fleet statistics. Planning decisions should use the upper forecast interval for capacity stress tests. Future work should add policy and COE variables only with a pre-specified causal or dynamic-regression design.

XIII. CONCLUSION

Among the evaluated models, Holt-Winters provided the most accurate 12-month holdout forecast and acceptable residual independence. The projected increase supports proactive infrastructure planning aligned with SDG 13.2, while the short history and the gap between registrations and emissions require conservative interpretation.

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